

A Contributors to Health Management for Persons Experiencing Homelessness: A Systematic Review

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Abstract

Objective: To examine published qualitative studies that can inform occupational therapists about potential intervention targets to improve health outcomes in homeless communities.

Methods: A search was conducted for pertinent studies in the following databases: PubMed, Ovid, and CINAHL. Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) guidelines were followed to extract relevant data. Only research published after 2011 in English-language peer-reviewed journals was included. Twelve of the 1,162 identified articles satisfied the requirements for inclusion.

Results: Individuals experiencing homelessness had higher levels of food insecurity, mental health conditions, and inadequate management of chronic conditions than those with a stable residence, all of which were successfully treated through occupational therapy.

Conclusion: The scope of practice of occupational therapists allows for holistic treatment approaches to address the biopsychosocial health of homeless individuals at all levels of need and suggests that occupational therapy intervention initiatives are appropriate to address occupational justice for this population

Key words: Food insufficiency, Health management, Homelessness, Mental health, Occupational Therapy, Substance abuse

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Received 16 November 2023; Revised 16 November 2023; Accepted 30 January 2024

I . Introduction

The number of people experiencing homelessness in the US has exceeded more than half a million and is steadily increasing (National Alliance to End Homelessness, 2023). The range of definitions of homelessness from this review spanned from unstable housing (Grazioli et al., 2015; Hernandez et al., 2019; Pendyal et al., 2021; Romaszko et al., 2017), short- and long-term stays at homeless shelters or hostels (Baggett et al., 2011, Brown et al., 2012, Groton et al., 2021, Manhapra et al., 2021, Salem et al., 2021), and persons living on the streets or in their vehicles (Fallaize et al., 2017). Homelessness results in persons having limited access to basic needs such as nutritious food, clean water, and a safe and consistent environment, and is a growing international social injustice and human rights problem (Marshall et al., 2021). Recent studies indicate that homeless individuals face many challenges in prioritizing immediate needs versus expenses related to managing chronic conditions. Homeless individuals often have had to choose between earning a few dollars or attending a doctor's visit and have made the choice that met their more pressing basic need: earning a few dollars (Pendyal et al., 2021).

Cardiovascular disease, mental illness, heart failure, diabetes mellitus, alcoholism, and substance abuse are prevalent among adults experiencing homelessness and may challenge self-management because of the transient nature of their residency status and lack of regular preventative care (Brown et al., 2012; Grazioli et al., 2015; Groton et al., 2021; Hernandez et al., 2019; Koh et al., 2016; Manhapra et al., 2021; Pendyal et al., 2021; Romaszko et al., 2017; Salem et al., 2021).

Homelessness studies also acknowledge related health and social concerns, such as high rates of victimization and justice system involvement, decreased functional capacity, and reduced life expectancy (Roy et al., 2017). Recurring themes such as instability, lack of routine, tradeoffs between basic needs and management of chronic conditions, and stigmatization by healthcare providers may continue to create barriers for the homeless population (Groton et al., 2021; Hernandez et al., 2019; Pendyal et al., 2021). Characterizing the lived experiences faced by homeless individuals will help to identify emerging themes and ground them into theories that may lead to improving the healthcare services provided to this vulnerable population.

Prior systematic reviews have addressed specific problems that hinder occupational engagement and optimal occupational performance in people experiencing homelessness, such as heavy drinking and malnutrition, and the role occupational therapy can play in transitioning individuals out of homelessness (Ijaz et al., 2018; Marshall et al., 2021). No holistic examination of the lived experiences of the homeless population has been conducted to better understand the factors limiting health management in this population. For example, a transient lifestyle may negatively impact engagement in occupations such as health management of chronic diseases, psychosocial mental functioning for social participation, feeding and nutrition, and participation in meaningful activities of daily living (American Occupational Therapy Association, 2020b).

This mixed methods systematic review was conducted to identify the potential role of occupational therapy in building health management skills for persons experiencing homelessness while managing their chronic healthcare conditions.

The purpose of this review is to compile and integrate themes from lived experiences to better understand the factors impacting health management within the homeless population. The researchers sought to identify the potential role occupational therapists can play in building physical and mental health management skills within the homeless population.

II. Methods

This systematic review protocol was registered with PROSPERO (CRD42021270797) on July 30, 2021 to identify the role of occupational therapy in building physical and mental health management skills in this population. The initial search identified a total of 1,126 articles. Of these articles, 12 met all inclusion criteria and were

included in the study. The majority of the articles were excluded due to age of study participants, the experience of acute illness or modifiable disease, and cognitive-related impairments inhibiting the ability to give consent. Details of the process for the inclusion and exclusion criteria are identified in Figure 1.

1. Search Strategy

A team of five graduate students (MA, SG, AH, CM, & NY) under the guidance of faculty members (DC & CH) conducted the literature searches in the following databases: Pubmed, Ovid, and CINAHL. The search dates were May 27th through June 25th, 2021. Search terms included “Homeless AND Nutrition”, (homeless persons) AND (self-care); (homeless person) AND (health disparity); “Homeless AND Occupational Therapy;” “Homeless

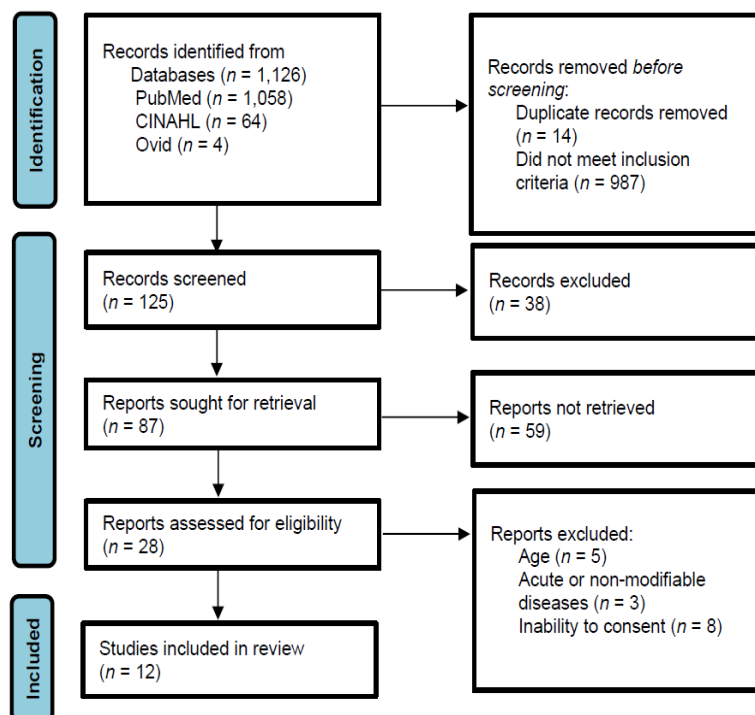


Figure 1. Flow Diagram

AND Resilience.” The search strings were adapted to the given topic across the databases. The systematic review was then conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (Page et al., 2021). Each article was independently reviewed by two students and later, notes were compared to assess eligibility. When disagreements occurred, the article was brought forth to the entire group for review where all members came to a consensus about the article’s final eligibility. Other questions regarding eligibility or risk of brought to the overseeing faculty members.

2. Selection Criteria

The following populations were excluded from this review: children and adolescents, adults experiencing acute diseases or non-modifiable diseases, and adults unable to provide consent due to cognitive impairments. Additional exclusion criteria included systematic reviews or expert opinions. The search was limited to studies published after 2011 and written in the English language. The inclusion criteria consisted of 1) studies published in peer-reviewed journals, 2) studies that produced qualitative and quantitative data, 3) studies examining health management concerns and problems faced by homeless persons, 4) studies that examined health management strategies and supports for homeless persons. Levels of evidence were evaluated with the guidelines recommended by the Centre for Evidence-Based Medicine (2009). An assessment tool by Higgins et al. (2016) guided the determination of the risk of bias for quantitative studies. When assessing qualitative studies, the Critical Appraisal Skills Programme (CASP)

Qualitative Studies Checklist was implemented (Critical Appraisal Skills Programme, 2018). The Flow Diagram produced below provides a comprehensive analysis of the authors’ approach to identifying included articles. It serves as a visual representation of the number of articles the search strings provided, what was assessed, and how the research was utilized (AOTA, 2020a).

3. Thematic Analysis

The 12 records selected for discussion in this review present data to discuss the factors related to health management skills for persons experiencing homelessness. Provided the heterogeneity of the studies selected, a narrative synthesis was composed to define themes to be designated as identified related factors. A complete synthesis is shown in Table 1 for the narrative findings in each included independent study. Findings were compared across studies and divided into appropriate categories of related factors. Themes were defined to identify factors related to health management that may be addressed by occupational therapy.

III. Results

The total number of participants included in the 12 studies (6 exclusively quantitative, 5 exclusively qualitative, 1 mixed method) was 5,453,497, from 18-69 years old, with sample population sizes ranging from 10 to 5,450,078 individuals. Participants had experienced some component of chronic disease, food insufficiency, mental health illness, and/or difficulty with health management. The studies were conducted

in Poland (Romaszko et al., 2017), Puerto Rico (Baggett et al., 2011), and the U.S. (Baggett et al., 2011; Brown et al., 2012; Chilton et al., 2015; Fallaize et al., 2017; Grazioli et al., 2015; Groton et al., 2021; Hernandez et al., 2019; Koh et al., 2016; Manhapa et al., 2021; Pendyal et al., 2021; Salem et al., 2021). Seven studies used quantitative methodologies (Baggett et al., 2011; Brown et al., 2012; Fallaize et al., 2017; Grazioli et al., 2015; Hernandez et al., 2019; Manhapa et al., 2021; Romaszko et al., 2017). The qualitative methodologies included grounded theory (Chilton et al., 2015; Grazioli et al., 2015; Groton et al., 2021; Pendyal et al., 2021), participatory action research (Koh et al., 2016; Salem et al., 2021), and phenomenology (Salem et al., 2021), using in-person interviews, medical background checks, and focus groups (Table 1).

Approaches to data collection varied among the studies but the commonly used approaches were: the US Household Food Security Survey Module (Chilton et al., 2015), the Alcohol and Substance-Use Frequency Assessment (Grazioli et al., 2015), the Michigan Alcoholism Screening Test (Romaszko et al., 2017), focus groups (Groton et al., 2021; Salem et al., 2021), questionnaires (Hernandez et al., 2019), and interviews (Koh et al., 2016; Pendyal et al., 2021; Romaszko et al., 2017). The mixed methods review also included one intervention study (Grazioli et al., 2015).

The identified outcomes across studies included how poor nutrition related to homelessness may contribute to chronic health conditions (Fallaize et al., 2017; Groton et al., 2021; Hernandez et al., 2019), how homelessness was a hindrance to the management of conditions following mental health diagnoses (Brown et al., 2012; Fallaize et al., 2017; Grazioli et al., 2015; Manhapa et al., 2021; Salem et al., 2021), and how homelessness

contributed to difficulty in managing and preventing acute and chronic illnesses secondary to limited or unattainable health care services (Brown et al., 2012; Salem et al., 2021).

1. Risk of Bias

All studies involving quantitative data were Level 3B and were determined to have a low risk of bias (National Heart Lung and Blood Institute, 2014) (Table 2). Researchers assessed risk of bias among the qualitative studies through use of the Critical Appraisal Skills Programme (CASP) checklist (CASP, 2018) (Table 3). All of the seven studies involving qualitative data had clear statements of the aims of the research and the findings and the value of the research was described. In addition, qualitative methodology and recruitment strategies were considered appropriate, and data were collected in a way that addressed the research issues. For five of the studies the relationship between the researchers and participants was not described so could not be adequately considered (Chilton et al., 2015; Grazioli et al., 2015; Groton et al., 2021; Koh et al., 2016; Salem et al., 2021). For four of the studies, ethical issues were not discussed so could not be taken into consideration (Grazioli et al., 2015; Groton et al., 2021; Koh et al., 2016; Salem et al., 2021). For three of the studies, it was not clear if data analyses were sufficiently rigorous (Chilton et al., 2015; Groton et al., 2021; Koh et al., 2016).

2. Synthesis

The results of the systematic review identified three major themes and various subthemes. The major themes were “food insufficiency,” “mental health,” and “management of chronic disease.”

Table 1. Evidence Table

Author/Year	Purpose Control Group Study Setting	Participants Inclusion Criteria	Level of Evidence Study Design / Risk of Bias (Quality Assessment)	Results (Including significance of findings)	Outcome Measures/How does it inform OT?
Baggett et al. (2011)	- Examined food insufficiency relationship to rates of hospitalization and emergency department visits. - No control group - Setting: 79 clinic sites operated by 30 grantees across US.	N = 966; In-person interviews. Inclusion: Individuals receiving fact-to-face services from 'Health Care of Homeless' (HCH) provider, and who received services at least once one year prior to the visit had surveys administered.	Level of Evidence - 3B Cohort Study Risk of Bias - Low	- 25% of homeless adults in the survey were food insufficient - Chronically homeless ($p = 0.01$) and traumatically victimized ($p = 0.001$) respondents were more likely to be food insufficient. - Significant relationships with hospitalization, psychiatric hospitalization, and high emergency department utilization - A significant association was found between food insufficient respondents (homeless) and psychiatric hospitalization compared to respondents who were food sufficient (general population).	- Food insufficiency was associated with increased odds of acute health services needed - Lack of primary care for homeless persons is a concern and should be matched by public health and policy initiatives to provide basic needs for the very poor.
	- To determine the prevalence of common geriatric syndromes in a sample of older homeless adults and compare the prevalence to those reported in the general older population. - Participants were assessed in ADL performance, falls in the past year, cognitive impairments, and other comorbidities. - No Control group - Setting: Outpatient clinical setting.	N = 247 homeless adults aged 50-69 yrs. recruited from 8 homeless shelters (emergency, transition, and day) in Boston, MA. 4:1 ratio of men to women. Inclusion criteria: Participants were > 50 years old, homeless, able to communicate in English, and able to provide written informed consent	Level of Evidence - 3B Cross-sectional Risk of Bias - Low	- 30% of patients had difficulty performing at least one activity of daily living. - 24.3% presented with cognitive impairments, 28.3% presented with executive function impairments, 16% presented with frailty, 39.8% had major depression, and 49.8% participants presented with urinary incontinence. - 29.7% and 30.0% self-reported visual and hearing impairments.	- OTs working with older adults from the homeless population should be aware that compared to older adults of similar characteristics, older adults who are homeless have been shown to have a higher rate of most geriatric syndromes. These factors will impact the plan of care for older homeless adults.
Brown et al. (2012)	- To investigate how the adverse childhood experiences (ACEs) of mothers affect their future food insecurities, this study conducted in-person interviews of quantitative (demographics, health status, economic circumstances, adverse childhood experiences), qualitative (quality and characteristics of childhood experiences with hunger, deprivation, abuse, and neglect) and open-ended questions.	N = 31: In-person interviews including quantitative measures of maternal and child health, ACEs, and food security	Level of Evidence - Qualitative study Risk of Bias - see CASP	- Strong relationship between exposure to adverse childhood experiences and household food insecurity between qualitative and quantitative results. - 52% of participants reported emotional and physical abuse & significance was found associated with very low food security at the household level and food insecurity at the child level.	- Deprivation among households with young children reporting very low household food security are related to mothers' own reports of exposure to ACEs.
Chilton et al. (2015)					

Table 1. Evidence Table (Continued)

Author/Year	Purpose Control Group Study Setting	Participants Inclusion Criteria	Level of Evidence Study Design / Risk of Bias (Quality Assessment)	Results (Including significance of findings)	Outcome Measures/How does it inform OT?
Falliaze et al. (2017)	- No control group. - Setting: Interviews conducted in participants' homes, with the exception of seven participants who requested to meet in the local library. Philadelphia, PA.	- Inclusion: Mothers of children < 6 yrs., mothers who reported low (multiple indications of food access problems and reduced diet quality) or very low household food security (food intake of at least one household member was reduced and eating patterns were disrupted during the year because the household lacked money and other resources for food), families requesting outreach services after participating in the ongoing Children's HealthWatch-Philadelphia study, English or Spanish-speaking	- Level of Evidence - 3B Case-control study Risk of Bias - Low	- 21 participants reported an ACE score that measured adverse childhood experiences of ≥ 4 and were more likely to report very low food security ($p = 0.021$). - Qualitative reports of exposure to violence, and neglect reflect experiences continuing to affect people into adulthood and as they are taking care of their own children. - Early exposure to trauma, violence, and neglect leads to an increase in fighting at school or quitting jobs for minor reasons, both of which impact financial stability in the future.	
	- Homeless and matched housed adults. - Case controls were used in this study. Setting: Age of 38 years. To be eligible for the homeless group individuals had to be "living it rough" on the street and eligibility for housed individuals was that they were in the initial stages of housing (supported living)	N = 57 homeless and 75 housed men and women - Inclusion: age 38 yrs., groups had to be "living it rough" on the street, and eligibility for housed individuals was that they were in the initial stages of housing (supported living)	- Level of Evidence - 3B Case-control study Risk of Bias - Low	- No significant difference was found in the BMI of homeless and housed adults. - 5-3% of the homeless adults were underweight while more than 50% of housed adults were overweight or obese (none were underweight). - 24% of homeless adults and 4% of housed adults reported mental illness diagnoses. - Dietary intake of homeless adults was higher in salts, alcohol, fiber, vitamin C, and fruit than housed individuals.	- Homeless adults had a higher risk for cardiovascular disease and higher incidence of anxiety and depression, as well as poorer dietary and nutritional intake. - Finding food and/or housing resources for their homeless clients would be essential.
Grazioli et al. (2015)	- Currently or chronically homeless adults with alcohol dependence. Researchers gave participants multiple safer-drinking strategies to implement throughout the duration of the study. - No control group. - Setting: Two community-based agencies on the forefront of harm reduction housing and service provision to chronically homeless people in a large city in the Pacific Northwest.	N = 31 Inclusion Criteria: Receiving services from 1 of 2 partnering agencies, being 21-65 years old, agreeing to use birth control, and fulfilling criteria for alcohol dependence according to the DSM-IV-TR.	- Level of Evidence - Mixed method Quantitative 3B Quasi-Experimental - Single group non-randomized pretest, posttest study Risk of Bias - Low	- Qualitative results showed buffering the effects of alcohol on the body to be the most common category for safer drinking habits. - Quantitative data showed no significant change in the number of safer-drinking strategies used by participants from week 0 to week 8.	- Implementation of harm-reduction interventions related to safer-drinking strategies may decrease alcohol-related harm and increase the amount of safer drinking strategies used by homeless individuals. - Increases awareness of the potential uses of harm-reduction interventions for the homeless population.

Table 1. Evidence Table (Continued)

Author/Year	Purpose Control Group Study Setting	Participants Inclusion Criteria	Level of Evidence Study Design / Risk of Bias (Quality Assessment)	Results (Including significance of findings)	Outcome Measures/How does it inform OT?
Groton et al. (2021)	Participants were placed into three focus groups that examined the barriers and facilitators of self-management of hypertension (HTN) while experiencing homelessness:	N = 18 participants (77% male; 2 focus groups had 5 people, one focus group had 8) Inclusion criteria: self-report of hypertension, had to be able to answer yes to "Are you currently on the medication you have been told is for treatment of high blood pressure?" and "Have you been told by a health provider that you have high blood pressure?"	Level of Evidence - 2 Qualitative Study Risk of Bias - see CASP	- Managing medications: most participants utilized local pharmacies: some utilized the ER - Healthy eating: most participants noted the importance of diet in managing HTN - Exercise: 8 participants noted that PCP suggested exercise to manage HTN; most felt they fulfilled this part of their healthcare - Social Support: 5 participants said they use their social networks to manage their HTN - Social support: 3 reported trying to reduce stress to keep blood pressure down, 1 reported getting up and walking away when others made him feel stressed	- Emphasizes the importance of healthcare professionals when addressing inequities in persons experiencing HTN and homelessness - Overall, most participants seemed to understand how to manage their HTN - Those with other comorbidities expressed the most difficulty with managing their HTN and other conditions.
	- No control group - Study setting: Southeastern Florida metropolitan county; participants from a well established hot-meal program in the community for persons experiencing homelessness				
Hernandez et al. (2019)	- To determine whether fruit and vegetable consumption (as a measure of diet quality) or emotional distress tolerance act as potential links between food insecurity and poor physical and mental health among adults experiencing homelessness.	N = 566 Participants: 648 participants were screened for study participation and 38 were disqualified due to lower literacy levels. Data were excluded if areas of interest were missing from the survey-- final sample included 566 homeless individuals.	Level of Evidence - 3B Cross-sectional study Risk of Bias - Low	Results: Significant relationships found between food insecurity and poor health, depressive feelings, and PTSD symptoms. Low levels of stress tolerance for people experiencing food insecurity contribute to poorer physical and mental health. Multiple factors play into the link between food access and emotional distress.	- Outcomes showed food insecurity affects physical and mental health outcomes - Increasing access to food can increase resilience to emotional stress and improve overall health outcomes. - OT's can play a role in health management and nutrition management through education and identification of resources.
	- Participants completed a questionnaire - No Control Group - Setting: not specified	Inclusion Criteria: minimum of 18 years of age, receiving shelter-based services (e.g., shelter, counseling) at the targeted shelters, and had a minimum 7th-grade English literacy level based on a score of 4 or higher on the Rapid Estimate Adult Literacy in Medicine-Short Form (REALM)			

Table 1. Evidence Table (Continued)

Author/Year	Purpose Control Group Study Setting	Participants Inclusion Criteria	Level of Evidence Study Design / Risk of Bias (Quality Assessment)	Results (Including significance of findings)	Outcome Measures/How does it inform OT?
Koh et al. (2016)	- Examined current practices, barriers, and ideas to improve the nutrition of homeless individuals - Shelters or soup kitchen food directors were interviewed. Open and closed-ended questions Setting: Shelters or soup kitchens.	N = 10: Conducted 60 min. in-person interviews with shelter or soup kitchen food directors from ten shelters in the Greater Boston area Inclusion: shelters in Greater Boston area	Level of Evidence - Qualitative study Risk of Bias - see CASP	- Homeless individuals suffer from a high prevalence of diseases related to poor diets. - This population can be given an opportunity to restore health and nutrition through education, policies in shelters, and soup kitchens. - Significant barriers to healthy foods for this population were budget constraints, followed by space constraints. - Lack of food education among shelter staff was a contribution to insufficient production of meals.	- Nutritionally related diseases are seen in homeless communities (i.e. hypertension, diabetes, and hypercholesterolemia) which decrease one's quality of life. - Need to shift the focus of shelters and soup kitchens to the quality of calories being consumed and government agencies implementing policies to assist in accomplishing these goals. - Alternatives to unhealthy foods can serve as substitutions.
	Compared the prevalence of opioid use disorder (OUD) with sociodemographic factors and clinical factors among homeless and non-homeless veterans. Control: veterans with permanent residences -Setting: over 1400 VHA facilities that provided healthcare services to eligible US military veterans in FY 2012	N = 5,450,078 veterans who received one or more inpatient or outpatient encounter(s) with the VHA at any facility during FY 2012 - Inclusion: Veterans with: a. > 1 inpatient or outpatient encounter coded with an ICD-9 diagnostic code indicating opioid abuse or opioid dependence b. > 1 clinic stop code or bed section DC code issued by specialized VHA homeless program b. receipt of services from a non-VA contract residential care program for homeless veterans c. a V-60 code indicating homelessness on > 1 outpatient encounter	Level of Evidence - 3B Case control Risk of Bias - Low	- OUD appears to be strongly related to homelessness among VHA clients.	Need for comprehensive strategies for VHA patients such as: - Money management - Health management - Medication management of OUD - Overdose prevention
Manhappa et al. (2021)	To identify the ways homelessness kept individuals living with heart failure (HF) from successfully self-managing the condition. - No control group - Setting: New Haven, CT.	N = 19 (11 were homeless at the time of the interviews) Inclusion: - 18 yo or older - experienced homelessness at any time following HF diagnosis	Level of Evidence - Qualitative study Risk of Bias - see CASP	Reasons that may hinder HF self-management in the homeless population include: - Difficulty incorporating successful HF self-management behaviors and routines due to lack of stability.	Need for: - Acknowledging that homelessness may be a barrier for HF self-management and working with clients to establish a daily routine to enable self-management through medication management

Table 1. Evidence Table (Continued)

Author/Year	Purpose Control Group Study Setting	Participants Inclusion Criteria	Level of Evidence Study Design / Risk of Bias (Quality Assessment)	Results (Including significance of findings)	Outcome Measures/How does it inform OT?
Romaszko et al. (2017)	- Aimed to better understand the homeless population living in Olsztyn, Poland, and the individual characteristics that define them.	- N = 98 - Mean age = 54.33 years old, mostly male, with an average of 10.19 years of schooling completed.	Level of Evidence - 3B Cohort study	- The constant tradeoff of choosing between prioritizing HF self-management vs. basic needs, Stigmatization by healthcare providers impacting HF care	- Tailor HF self-management education that is attuned to a homeless client's specific life challenges/ choices/tradeoffs - Shared decision-making with the client that incorporates social determinants of health (i.e. housing) that enables the client to share their values, goals, and barriers to enable engagement in HF self-management
	- No control group Setting: Olsztyn, Poland. Olsztyn has a population of 175,000, 153 of whom are homeless	Inclusion criteria: homeless people who either checked in for the night or appeared at the Sabina Kusznirow Memorial Shelter for Homeless People at some point during the 12-month study. Participation was voluntary. Those who were visually impaired, or who could not read, were read the information by the investigator.	Risk of Bias - Low	- Mean BMI = 23.68 kg/m ² , 84.69% were smokers, and 81.69% had access to a general practitioner and had their medical treatment covered through social security. - 58 participants were willing to work for pay. Those willing to work for pay completed more years of school than those unwilling to work. - 66.33% used overnight homeless shelters, 27% stayed at homeless hostels, and 6.12% resided in public places, but these categories were not mutually exclusive. - Alcohol consumption and age were statistically significantly correlated with lower total alcohol consumption in older participants. Hazardous drinking was declared by 26 subjects and correlated with increased self-harm. - Almost a third of participants occasionally went hungry, which seemed to increase with age, but not income or education level. The primary source of food for 89.80% of the study population came from meals provided by welfare services. 80.61% of them spent more than half of their income on food, and 84.69% stated they ate at least 2 meals per day.	- Services that provide food, clothing, and medicine, as opposed to financial services, are more effective at satisfying the needs of the homeless population. - This study informs OT because it describes individual experiences had by homeless people. This allows OTs to better understand the homeless population and the types of obstacles they face. - Knowing that more than half of the participants were willing to work for pay, informs OTs that they would benefit from a job training and placement program.

Table 1. Evidence Table (Continued)

Author/Year	Purpose Control Group Study Setting	Participants Inclusion Criteria	Level of Evidence Design / Risk of Bias (Quality Assessment)	Results (Including significance of findings)	Outcome Measures/How does it inform OT?
Salem et al. (2021)	Examined perceptions of barriers and facilitators experienced by formerly incarcerated homeless women during reentry into the community. No control group Setting: Either one of two residential drug treatment programs in Los Angeles, California	N = 18: Formerly incarcerated, homeless women experiencing substance abuse, ages 18-65 years - Three focus groups with four to seven participants in each focus group were used to collect data - Inclusion: Homeless women exiting jail or prison, currently on parole and/or probation, charged with a drug-related offenses	Level of Evidence - Qualitative study Risk of Bias - see CASP	- Women lacked knowledge about resources (dental and mental healthcare, health insurance). - Inaccessibility to health insurance is problematic as formerly incarcerated individuals have increased chronic health conditions. - Participants had a strong desire to seek housing and independence, though available housing options were limited.	- Interventions facilitated to prepare for release are important during one's time spent in prison. - Needs included strengthening relationships between family and friends, along with building problem-solving skills, and job preparation. - Necessity of preparing jailed persons for life following incarceration.

ACEs: Adverse childhood events, CASP: Critical appraisal skills programme, HF: Heart failure, HTN: Hypertension, OT: Occupational therapy, OUD: Opioid use disorder, SM: Self-managing, VHA: Veterans health administration.
ADL: Activities of daily living, BMI: Body mass index, DC: Discharge, DSM-IV-TR: Diagnostic and statistical manual of mental disorders, version 4, text revised, ER: emergency room, FY: Fiscal year, ICD-9: International classification of disorders, version 9, PCP: Primary care provider, PTSD: Post traumatic stress disorder, VA: Veteran's administration
Centre for Evidence-Based Medicine (2009).

Table 2. Risk of Bias Before-After (Pre-Post) Studies with No Control Group

Citation	Study question or objective clear	Eligibility or selection criteria clearly described	Participants representative of real-world patients	All eligible participants enrolled	Sample size appropriate for confidence in findings	Intervention clearly described and delivered consistently	Outcome measures pre-specified, valid/reliable, and assessed consistently	Assessors blinded to participant exposure to intervention	Loss to follow-up after baseline 20% or less	Statistical methods examine changes in outcome measures from before to after intervention	Outcome measures were collected multiple times before and after intervention	Overall risk of bias assessment (low, moderate, high risk)
Baggett et al. (2011)	Y	Y	N	Y	Y	NA	Y	NA	Y	NA	NA	L
Brown et al. (2012)	Y	Y	Y	N	Y	NA	N	NA	Y	NA	NA	L
Fallaize et al. (2017)	Y	N	Y	Y	N	Y	Y	NA	Y	NA	NA	L
Grazioli et al. (2015)	Y	Y	Y	Y	N	Y	Y	NA	Y	Y	N	L
Hernandez et al. (2019)	Y	Y	Y	Y	Y	NA	Y	NA	Y	NA	NA	L
Manhapra et al. (2021)	Y	Y	Y	Y	Y	NA	Y	NA	Y	NA	NA	L
Romaszko et al. (2017)	Y	Y	Y	Y	Y	NA	Y	NA	Y	NA	NA	L

Y: yes, N: no, NA: Not available

Scoring for overall risk of bias assessment is as follows: 0-3 N: Low risk of bias (L), 4-8 N: Moderate risk of bias (M), 9-11 N: High risk of bias (H).
National Heart Lung and Blood Institute. (2014).

Table 3. CASP Qualitative Checklist

Author	Was there a clear statement of the aims of the research?	Is the qualitative methodology appropriate?	Was the research design appropriate to address the aims of the research?	Was the recruitment strategy appropriate?	Was the data collected in a way that addressed the research issue?	Has the relationship between research and participants been adequately considered?	Have ethical issues been taken into consideration?	Was the data analysis sufficiently rigorous?	Clear statement of findings?	How valuable is the research?
Chilton et al. (2015)	Yes (1)	Yes (1)	Yes (1)	Yes (1)	Yes (1)	Can't tell	Yes (1)	Can't tell	Yes (1)	Valuable
Grazioli et al. (2015)	Yes (1)	Yes (1)	Can't tell	Yes (1)	Yes (1)	Can't tell	Can't tell	Yes (1)	Yes (1)	Valuable
Groton et al. (2021)	Yes (1)	Yes (1)	Yes (1)	Yes (1)	Yes (1)	Can't tell	No	No	Yes (1)	Valuable
Koh et al. (2016)	Yes (1)	Yes (1)	Yes (1)	Yes (1)	Yes (1)	No	No	No	Yes (1)	Valuable
Pendyal et al. (2021)	Yes (1)	Yes (1)	Yes (1)	Yes (1)	Yes (1)	Yes (1)	Yes (1)	Yes (1)	Yes (1)	Valuable
Salem et al. (2021)	Yes (1)	Yes (1)	Yes (1)	Yes (1)	Yes (1)	Can't tell	Can't tell	Yes (1)	Yes (1)	Valuable

Key: Yes (1), Can't Tell, No.

CASP: Critical Appraisal Skills Programme.

CASP (2018).

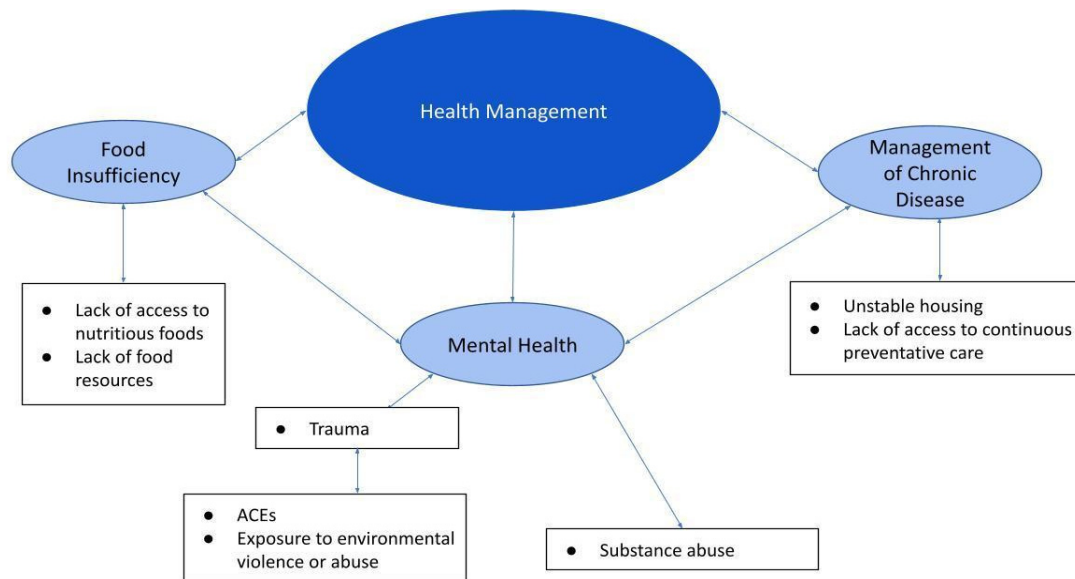


Figure 2. Concept Map

Themes and subthemes are depicted in Figure 2.

3. Theme 1: Food Insufficiency

Food insufficiency was common among much of the research and indicates how homeless individuals experienced inadequate dietary intake and the implications on their health and well-being. A second-order theme of lack of access to nutritious food resources was identified. This second-order theme will be further discussed.

The impact of food insufficiency was evident in individuals experiencing homelessness (Baggett et al., 2011; Chilton et al., 2015; Fallaize et al., 2017; Hernandez et al., 2019; Koh et al., 2016). Fallaize et al. (2017) found that 27% of the homeless individuals sampled described not having enough to eat, while 38% reported consuming less than, or equal to, one meal per day. Access to consistent meals, such as from a shelter or soup kitchen, was found to have a positive effect on emotional well-being (Hernandez

et al., 2019). This is important because food insufficiency was listed as a primary reason for homeless people seeking healthcare services. Individuals experiencing food insufficiency were more likely to be hospitalized, in both standard hospitals as well as psychiatric hospitals, and to seek emergency services (Baggett et al., 2011). Baggett et al. (2011) found that one in every eight participants reported hunger as the reason for seeking health care services.

Lack of access to nutritious food resources.

Gaining access to nutritious foods has been shown to be difficult for persons experiencing homelessness. Fallaize et al. (2017) concluded that homeless adults have a higher intake of fats, saturated fats, salt, and alcohol, and a lower intake of fruits, vitamin C, and dietary fiber than housed individuals. The study further found that although the diets of homeless individuals were similar in caloric intake to those of housed individuals, most of the calories were consumed in one meal, leaving the individual to report experiencing hunger. Likewise, poor nutritional

intake was shown to contribute to a higher risk for cardiovascular disease (CVD) and other chronic health conditions. Studies also found a significant relationship between food insecurity and increased risk of CVD and between the increased impact of anxiety, depression, and post-traumatic stress disorder (PTSD) symptoms (Fallaize et al., 2017; Groton et al., 2021; Hernandez et al., 2019).

Currently, no national nutritional standards have been established for food served in shelters and soup kitchens (Koh et al., 2016). Koh et al. (2016) found that some of the most significant barriers to healthy food for the homeless population were budget constraints, as well as space constraints for food kitchens. Another study found that homeless individuals residing in hostels had higher consumption of saturated fats than homeless individuals living on the street (Fallaize et al., 2017).

4. Theme 2: Mental Health

Mental health outcomes describe how the homeless population experience mental health and the impact it has on their transient lifestyle. They provide outcomes illustrating how mental illness may contribute to food insufficiency and negative physical health outcomes. This systematic review categorized themes into first, second, and third-order themes. A thorough review of the selected articles revealed common themes in multiple studies. The first-order themes, or the core themes branch into second- and third-order themes to further explain the relationships found within the core themes. The two second-order themes to be considered include (a) substance abuse and (b) trauma. A third-order theme emerged from the data linking trauma to food

insecurity: (a) adverse childhood experiences (ACEs) and (b) exposure to environmental violence or abuse.

Many of the studies in our review discussed the effects of mental illness on persons experiencing homelessness (Brown et al., 2012; Fallaize et al., 2017; Grazioli et al., 2015; Manhapra et al., 2021; Salem et al., 2021). Salem et al. (2021) studied the impact of mental illness on formerly incarcerated homeless women. Participants self-reported a variety of conditions including depression, bipolar disorder, and anxiety and a lack of adequate access and support from mental healthcare providers, family, or social services. Manhapra et al. (2021) found that mental health diagnoses were more prevalent for homeless veterans than non-homeless veterans. Similarly, persons experiencing homelessness were 41.3% more likely to be diagnosed with a mental health condition than non-homeless individuals. Overall, homeless individuals experienced depression, anxiety, bipolar disorder, PTSD, personality disorder, dysthymia, and schizophrenia at a higher rate than non-homeless individuals (Brown et al., 2012; Fallaize et al., 2017; Grazioli et al., 2015; Manhapra et al., 2021; Salem et al., 2021).

Substance abuse. Several studies found a large population of homeless individuals experiencing substance use disorders. In a study of 98 homeless participants, 78.57% of the participants were alcohol dependent and 84.69% were cigarette smokers (Romaszko et al., 2017). Romaszko et al. (2017) found the most common reason for resignation from social welfare was due to alcohol dependence. Moreover, unemployed individuals were spending a substantial proportion of their income on alcohol and cigarettes. Manhapra et al. (2021) found homeless veterans to be at a much higher risk for substance use

disorders than non-homeless individuals, and Grazioli et al. (2015) identified factors associated with reducing alcohol-related harm such as safer-drinking strategies.

Trauma. Recent trauma and past childhood trauma were both identified as contributors to mental health problems and subsequent health management difficulties. Exposure to adverse childhood experiences (ACEs), abuse, and very low food security can negatively influence individuals' engagement in desired occupations throughout their lifespan (Baggett et al., 2011; Chilton et al., 2015). Research has identified how difficulties occurring in childhood and adolescence have a significant impact on one's behaviors, choices, and relationships. ACEs include experiences such as violence, abuse, and neglect resulting in heightened states of arousal, withdrawal, and/or social isolation (Chilton et al., 2015). These experiences have been shown to lead to decreased safety, security, familial reconciliation and parenting and job stability in adulthood (Salem et al., 2021). Additionally, Chilton et al. (2015) found that families experiencing food insecurities had a higher prevalence of ACEs. Experiencing ACEs was associated with higher levels of depression and emotional disturbances directly affecting education and employment.

Contributing environmental exposures. Chilton et al. (2015) found that 68% of homeless participants reported four or more ACEs. Additionally, mothers who reported emotional and physical abuse significantly reported having very low food security ($p = 0.021$ & $p = 0.032$, respectively). Reports of participants of early exposure to violence, trauma, and neglect continue to affect experiences throughout adulthood and when taking care of their own children. The

impact of exposure to ACEs and violence influences emotional health, school performance, ability to maintain employment, and financial stability in the future. These markers of vulnerability continue to be addressed as both Baggett et al. (2011) and Chilton et al. (2015) acknowledge the significant relationship between traumatically victimized individuals and food insufficiency.

5. Theme 3: Management of Chronic Disease

Management of chronic disease explains how participants experiencing homelessness frequently faced a variety of barriers in managing chronic disease. Two second-order themes emerged from the data: (a) unstable housing and (b) lack of access to continuous preventative care.

Unstable housing. A common theme among the studies reviewed was the lack of stable housing for people experiencing homelessness due to the shortage of affordable housing options (Pendyal et al., 2021). Many found it difficult to manage their medication routines and medical care due to their unstable housing circumstances (Groton et al., 2021; Pendyal et al., 2021). In addition, some reported being victims of medication theft or having misplaced their medications because they did not have a secure location to leave their belongings (Groton et al., 2021). These individuals resorted to leaving their medication in a storage unit, in their vehicle, or with a family member for safekeeping.

Lack of access to continuous preventative care. In addition to unstable housing, lack of access to preventative care was shown to be another factor influencing the health of the homeless population. Older adults experiencing homelessness were shown to have a higher rate of geriatric

syndromes and hospitalization than the average older adult (Brown et al., 2012). Geriatric Syndromes can include functional, cognitive, visual, and hearing impairments as well as fragility, urinary incontinence, and depression. Management and prevention of symptoms related to these syndromes are directly related to access to healthcare services. Due to barriers restricting the provision of healthcare services, an increase in hospital admittance, hospital stay, cost of care, and overall health risk was seen in the homeless population (Brown et al., 2012). The article by Salem et al. (2021) adds that previously incarcerated homeless women experience a difficult time obtaining healthcare services due to long wait times and the inability to receive medication to manage their medical conditions.

IV. Discussion

In this review, the overall theme was the lack of housing stability and how it ultimately impacts access to early and preventative intervention. Lack of stable housing negatively impacts the ability to perform basic self-care for these individuals, who are frequently faced with the tradeoff between prioritizing health self-management and meeting their basic needs (Pendyal et al., 2021). The transient lifestyle experienced by individuals experiencing homelessness leaves this group of individuals vulnerable with decreased ability to self-manage their mental and physical health.

1. Theme 1: Food Insufficiency

Implications for addressing food insufficiency range from having access to enough food to having access to more nutritious food. A heightened prevalence of diseases exists within the homeless community related to food insecurity, such as hypertension, diabetes, and hypercholesterolemia; however, there have been few public health efforts on improving nutrition (Fallaize et al., 2017; Koh et al., 2016). The prevalence of food insufficiency in the homeless population in the US is sixfold greater than in the general US population (Koh et al., 2016). The potential for improvement of these circumstances exists that could start in shelters and soup kitchens. For instance, a food budget of \$300 U.S. dollars (USD) per week in one shelter in Massachusetts allowed substitution of whole grain bread for white bread, 1% milk for whole milk, and nuts for pastries. Improving the quality of foods, rather than continuing to donate foods of poor nutritional value and increasing the opportunities for nutritional education within shelters and soup kitchens may lead to the likelihood of the homeless population eating healthier foods. A similar study conducted by Fallaize et al. (2017) found that a higher risk of CVD, higher incidence rates of anxiety and depression symptoms, and poorer dietary and nutritional intake was seen among homeless individuals compared to housed adults. High consumption of meat, fats, and oils, as well as alcoholic beverages, coupled with significantly lower intakes of fruits, vegetables, and nuts, were found to be the nutritional characteristics of persons experiencing homelessness. In addition, dietary deficiencies negatively impact the already heightened prevalence of diseases within the

homeless community. Pendyal et al. (2021) emphasized the gravity of the disease that homeless individuals face, as mortality rates from CVD are two to three times greater among people experiencing homelessness than that of the general population.

2. Theme 2: Mental Health

Higher rates of mental illness in the homeless population suggest the potential for reduced ability to manage health. Interventions for individuals with mental illness could be beneficial to supporting skills to improve health management, both through individual and community-oriented programs.

Substance abuse. In addition to malnutrition, individuals living in homelessness are more susceptible to substance abuse and the use of illicit substances. Romaskzo et al. (2017) found that 78.57% of the 98 homeless participants were alcohol-dependent, and the majority spent their funds on alcohol and cigarettes. Such findings indicate the need for intervention and support systems, which minimize the risk of promoting and furthering alcohol and nicotine dependence. Manhapra et al. (2021) found a substantially higher prevalence of opioid use disorder among homeless veterans when compared to other Veteran Health Administration (VHA) participants, further highlighting the need for interventions in providing homeless individuals with resources and support for their mental health and socio-economic vulnerabilities. Similarly, the implications of trauma exposures seen as ACEs among homeless mothers were identified as indicators for substance abuse in the homeless population (Chilton et al., 2015). These findings suggest a need for resources to aid this vulnerable population in overcoming their

traumatic pasts and prevent further substance abuse and dependency. Grazioli et al. (2015) determined homeless individuals experiencing substance abuse to be open to learning modifiable behaviors to help reduce the risks associated with substance use. This presents an opportunity for occupational therapists to intervene by promoting coping strategies that may prevent substance use disorders and alcoholism.

Trauma and contributing environmental exposures. Trauma from recent events and from ACEs may be addressed individually through coping and resilience training strategies or through community programs to address those skills. Contributing environmental exposures may be addressed through improved housing opportunities which might result from individual life skills training for independent living or through community housing projects.

3. Theme 3: Management of Chronic Disease

Unstable housing. Unstable housing may be related to reduced ability to perform independent living tasks or to participate in employment. These issues may be addressed by programs to develop skills such as budgeting and management of household responsibilities or employment preparation or coaching. These supports could be developed for individuals or for community groups.

Lack of access to continuous preventative care. This may be addressed individually through training programs for people who struggle with independent living or community mobility skills or through group or community interventions to increase availability of community sites of providers.

Implications for Occupational Therapy

The themes discovered among the studies reflect areas in which interventions by occupational therapists can impact their homeless clients. Health management, nutrition management, primary care, engaging in community programs, advocating for adequate housing resources and substance abuse interventions are all within the scope of the occupational therapy profession (AOTA, 2020b).

Because homelessness is complex and includes an abundance of needs to be addressed a holistic approach is essential. Research has demonstrated the relevance and importance of occupational therapists' promotion of healthy routines and environmental modifications for the homeless population (Van Oss et al., 2020). By focusing on these areas through interventions, occupational therapists can have a powerful impact on the quality of life of people experiencing homelessness. The *Occupational Therapy Practice Framework: Domain and Process, Fourth Edition* (AOTA, 2020b) outlines how the occupational therapy scope of practice addresses activities of daily living such as personal hygiene and functional mobility. The framework further identifies the occupations of health management, medication management, physical activity, nutrition management, rest and sleep, financial management, community mobility, and social and community participation. Each of these occupations are greatly impacted by being homeless and their limitations reduce the occupational justice or access to and participation in the full range of meaningful and enriching occupations afforded to this population. Occupational therapists can aid in health management by guiding the individual to create and maintain a health management routine tailored to their specific needs. As nutrition management is not innately understood by

individuals experiencing homelessness, it creates a need for proper education and planning to be able to understand food as not only a source of energy but also as supplemental support for the physiological responses of the human body. Occupational therapists can enable their clients in better meal planning by setting small, manageable goals for improving their nutritional intake or in community programs supporting availability of more nutritional foods, such as community gardens. Support for finding and advocating for housing resources for homeless individuals is also among the occupations addressed by occupational therapists. Occupational therapists approach substance abuse holistically, addressing underlying unmet needs and/or coping strategies by facilitating engagement in everyday activities that promote health.

Occupational therapy also supports social justice and addressing social inequities (AOTA, 2020b). The *OTPF-4* defines occupational justice as the recognition of inclusive participation in routine meaningful occupations (AOTA, 2020b). This concept considers the instability of habits, roles, and routines of a homeless individual, which may result in a disruption of chronic disease management; it takes a multidimensional approach in determining treatments for the most potent intervention. Increasing evidence has demonstrated the impact of occupational therapy in the transition out of homelessness. This systematic review serves as a resource in evaluating the occupational barriers experienced by individuals facing homelessness and the potential role of occupational therapy intervention.

Strengths and Limitations

This review provides the groundwork for development of future occupational therapy interventions addressing critical needs of homeless

individuals. The mix of methodologies provided quantitative information about specific issues related to healthcare management by homeless individuals along with qualitative explanations about the underlying contributors to these issues.

Because qualitative studies delve into the lived experiences and personal perspectives of participants, researchers can gain a richer and stronger understanding of the factors impacting the lives of the homeless individuals and pull themes from the research to better understand this population (Kim et al., 2017). Careful synthesis of the research revealed a substantial amount of information examining the experiences of homeless individuals and the difficulties they face that could be addressed by occupational therapists. Multiple themes regarding needs within the homeless population were clearly identifiable.

Recruitment of participants was limited in most studies to localized urban areas and findings may not be transferable to all individuals. The vast majority of studies were conducted in the U.S., so may not reflect the experiences of homeless individuals in other countries. Additionally, the frequent changes in housing experienced by homeless individuals make it difficult to track participants across time. Settings for recruitment of participants are also limited in number, which restricts the overall population sample. Only two studies included sample populations outside of community-based homeless shelters, which included veteran associations and drug treatment programs (Manhapra et al., 2021; Salem et al., 2021); these samples may have a limited variety of lived experiences of individuals experiencing homelessness. Last, there exists a broad spectrum of needs impacting the homeless individuals, but few quantitative studies have been conducted to

substantiate findings.

Future Recommendations

Future research is recommended to further examine the overall health and wellness of homeless individuals and interventions to address their health management. Baggett et al. (2011) determined the importance of further studies examining the impact of improved food services on healthcare utilization for this population. Due to smaller sample sizes in many of the studies, further research is needed to determine the prevalence of hypertension and cardiovascular disease and the effects of adverse health conditions affecting homeless individuals, and outcomes of interventions to address health management (Groton et al., 2021; Romaszko et al., 2017). Additional studies are needed to investigate how occupational therapy can address the need for community reintegration by meeting basic needs, enabling autonomous health management, instructing in more effective coping mechanisms, and teaching and supporting self-advocacy.

Future research should also examine the impact of holistic approach options to the treatment of individuals experiencing homelessness. Current research studies have examined single-faceted approaches to treatment, such as addressing hypertension (Groton et al., 2021) and food insufficiency (Baggett et al., 2011; Chilton et al., 2015; Hernandez et al., 2019; Koh et al., 2016) rather than addressing treatment from a holistic point of view. This gap indicates a need for the development of interventions that employ more integrated holistic approaches to intervention.

Occupational Therapy Implications

This systematic review summarizes the limited evidence for areas of occupation for individuals experiencing homelessness who might benefit

from occupational therapy interventions. The implication for occupational therapy practice and future research in these areas are as follows:

- Develop occupational therapists' role and examine effectiveness in the treatment of homeless individuals by furthering research on interventions focusing on nutrition management, chronic disease management, teaching coping strategies, as well as training in activities oriented toward taking care of one's own body and activities that support daily life within the home and community (AOTA, 2020b).
- Implement client-centered interventions addressing mental health illness through use of creative expression, emotional regulation, coping strategies, increasing self-awareness, implementation of leisure and self-care opportunities, self-advocacy, and connecting to community resources for individuals experiencing homelessness.
- Increase the use of preventative healthcare for homeless individuals, when managing chronic disease residual impacts, including hypertension, diabetes, cardiovascular disease, and preventing the onset of acute illness, resulting from food insufficiencies, malnutrition, and infections.
- Expand the role of occupational therapy in primary care to address the prevention of chronic conditions for homeless individuals such as hypertension, cardiovascular disease, diabetes, etc. (AOTA, 2020b; Baggett et al., 2011; Wang et al., 2018)
- Future research should address OT's role in building autonomy and addressing needs during the homeless individual's transition into independent living (AOTA, 2020b; Fallaize et al., 2017).

V. Conclusion

Individuals experiencing homelessness are vulnerable to many adversities due to the transient and unsupportive nature of this lifestyle. Food insufficiency, mental health problems, and inadequate management of chronic diseases are identified as primary factors hindering the well-being of those without a stable residence. This combination of factors suggests a need for a holistic approach to the treatment of this transient population, rather than a single-faceted approach. Findings from this systematic review suggest a compelling need for expanding evidenced-based occupational interventions for the homeless population to improve health outcomes and address social justice in this population.

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